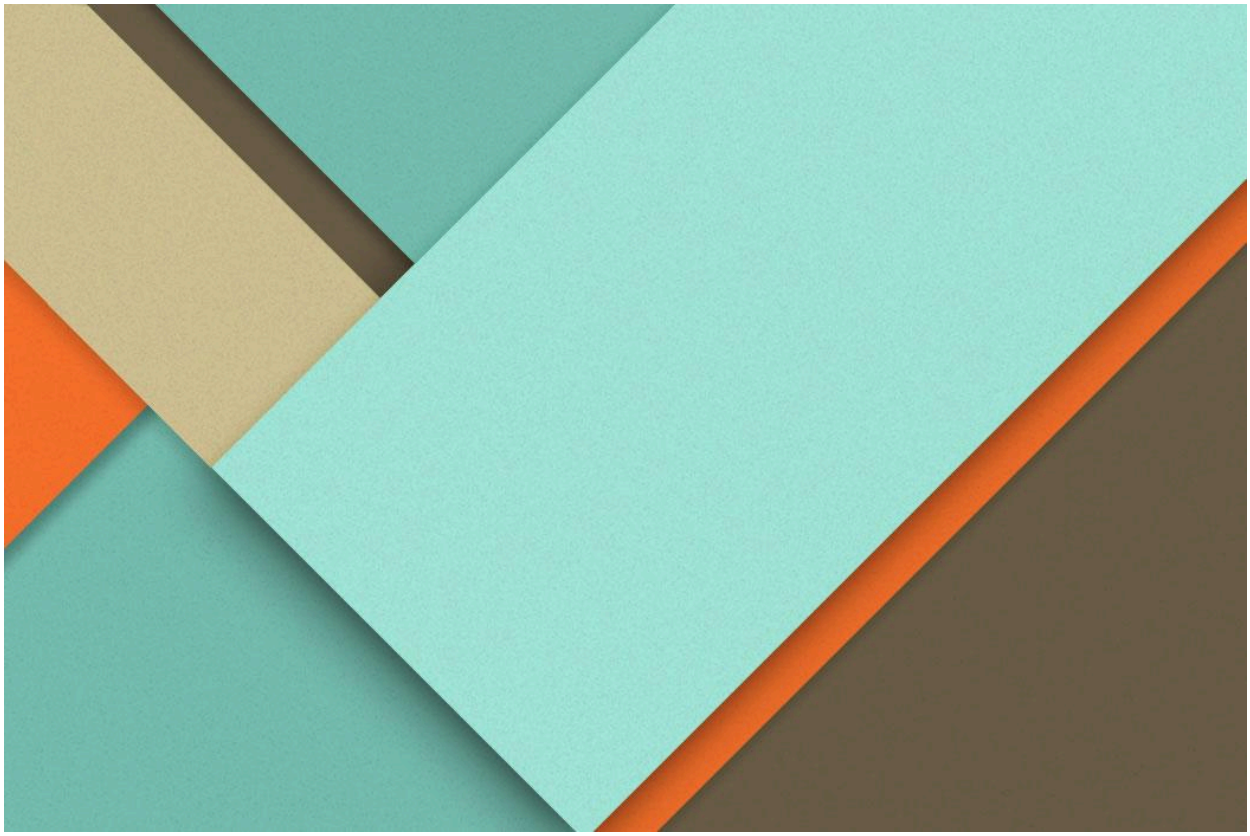

Loan Eligibility Prediction

—

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Machine Learning / Data Science Project

Overview

In today's rapidly evolving financial landscape, the demand for loans has increased significantly as individuals and businesses rely on credit to meet personal, educational, housing, and entrepreneurial needs. Financial institutions receive a large volume of loan applications daily, making it challenging to evaluate each applicant quickly while maintaining accuracy and fairness. As a result, there is a growing need for intelligent tools that can assist in the preliminary screening of loan applicants.

The Quick Loan Eligibility Predictor is designed to address this challenge by providing a fast and data-driven method to estimate whether a loan applicant is likely to qualify for a loan. Using historical financial data and key applicant attributes such as income, employment status, credit history, and property area, the system can generate a prediction that helps financial institutions quickly assess the potential eligibility of an applicant.

Goal and Audience

The goal of this project is to build a predictor that can classify loan approval status (Loan_Status). Financial services companies could benefit from utilizing this predictor to quickly screen applications and minimize applications that require in depth manual review and approval.

Data Source

Existing data was sourced from Kaggle with the following data points:

Column	Description	Example
Customer_ID	Unique identifier for each loan applicant	569
Gender	Gender of the applicant	Male / Female

Married	Marital status of the applicant	Yes / No
Dependents	Number of dependents	0, 1, 2, 3
Education	Education level of the applicant	Graduate / Not Graduate
Self_Employed	Whether the applicant is self-employed	Yes / No
Applicant_Income	Applicant's monthly income	5000
Coapplicant_Income	Coapplicant's monthly income	1500
Loan_Amount	Loan amount requested (in thousands)	128
Loan_Amount_Term	Loan repayment term (in months)	360
Credit_History	Credit history meets lending criteria (1 = Yes, 0 = No)	1
Property_Area	Type of property area	Urban / Semiurban / Rural
Loan_Status	Loan approved or not (target variable)	Y / N

The **target variable** for prediction is:

- Loan_Status

EDA

I. Feature Engineering

The original dataset was already in relatively good condition and required minimal cleaning.

However, several additional features were engineered to better reflect financial factors commonly used during loan application screening:

- Total_Income
  - Applicant_Income + Coapplicant_Income
- Monthly_Payment
  - Full_Loan_Amount (in thousands) / Loan_Amount_Term
- Debt_To_Income_Ratio
  - Monthly_Payment / Total_Income

These additional features help provide stronger predictive signals for the models.

II. Data Cleaning and Processing

The following data processing was done for the data set:

- One hot encoding for categorical features such as Property_Area
- Binary categorical variables mapped to numeric values (0/1)
  - Gender
  - Education
  - Married
  - Loan_Status
- Feature scaling applied to numeric columns before modeling
- Two anomalous records removed after investigation during analysis

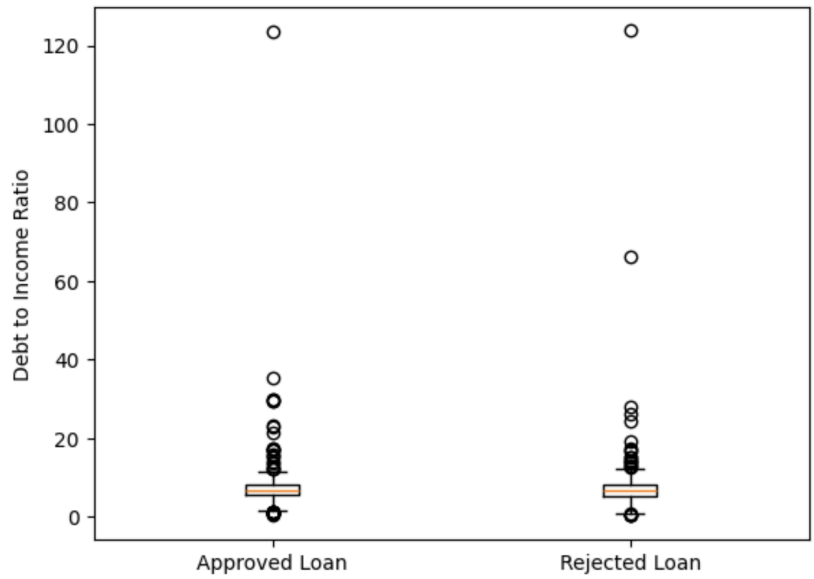
III. Data Analysis

Initial investigation began by examining distributions across features.

Analysis revealed that income alone does not determine loan approval, which motivated the creation of the Debt_To_Income_Ratio feature.

Outlier Investigation

Assuming Debt_To_Income_Ratio and Credit_History are strong predictors of loan approval, the data was examined by slicing approval status.



An anomaly was discovered:

- Record ID 498 showed:
  - Debt-to-income ratio > 100%
  - No credit history
  - Yet the loan was approved

This is unusual because a loan with a monthly payment exceeding total income would rarely be approved.

Further verification showed:

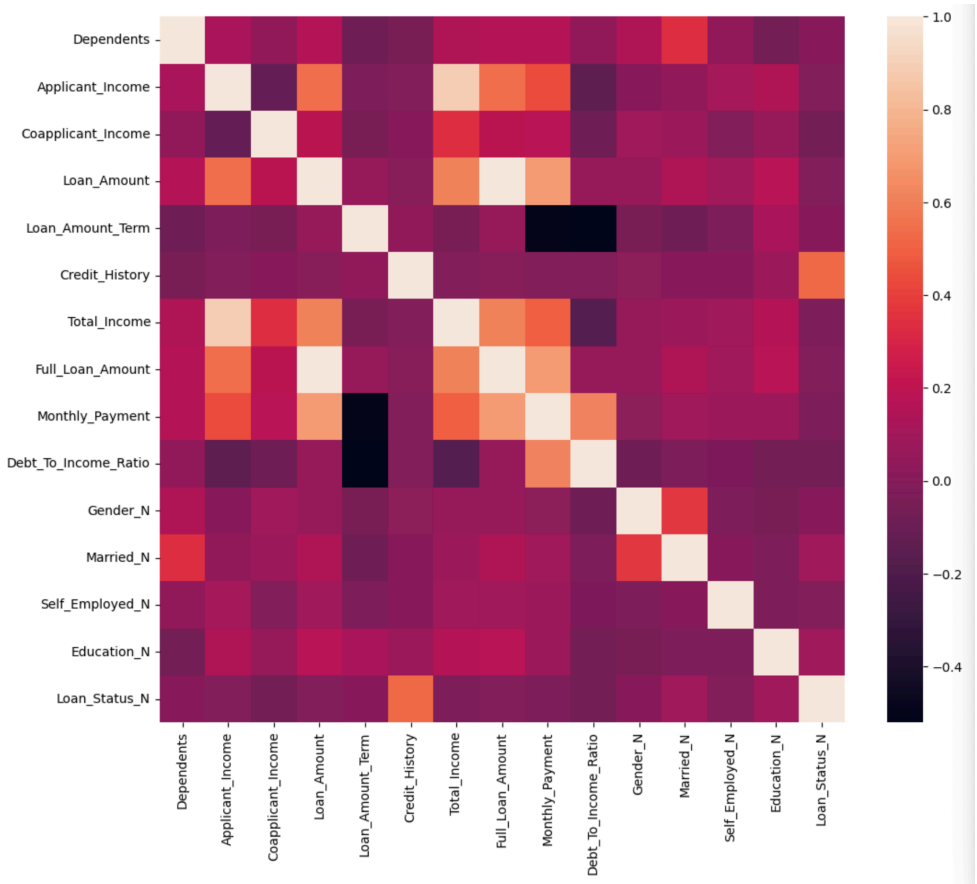
- Approved loans without credit history typically had Debt_To_Income_Ratio < 11%

Customer_ID	Total_Income	Full_Loan_Amount	Loan_Amount_Term	Monthly_Payment	Debt_To_Income_Ratio	Loan_Status	Credit_History
498	7482.0	111000	12	9250.00	123.63	Y	0
454	4949.0	90000	180	500.00	10.10	Y	0
202	4923.0	166000	360	461.11	9.37	Y	0
319	3541.0	112000	360	311.11	8.79	Y	0
156	39999.0	600000	180	3333.33	8.33	Y	0
327	4917.0	130000	360	361.11	7.34	Y	0
268	6144.0	160000	360	444.44	7.23	Y	0
528	6715.0	161000	360	447.22	6.66	Y	0
199	6232.0	113000	360	313.89	5.04	Y	0
123	11117.0	137000	360	380.56	3.42	Y	0

Because record 498 violated this pattern, it was classified as bad data and removed from the dataset.

Correlation Findings

Correlation matrix showed that Credit_History is strongly correlated with Loan_Status, indicating it is one of the most influential predictors in the dataset.



Modeling

The original dataset is imbalanced, with a larger number of observations representing loan rejections (69%) than approvals (31%). Because of this imbalance, the target variable (Loan_Status) was stratified during the train–test split to ensure that both training and testing datasets preserved the original class distribution.

Several machine learning models were evaluated to determine which approach provided the most reliable predictions for loan approval.

I. Logistic Regression

Logistic Regression was implemented as a baseline classification model. To optimize

model performance, GridSearchCV was used to perform hyperparameter tuning.

The search identified the following best-performing configuration:

- C = 10
- penalty = 'l1'
- solver = 'liblinear'

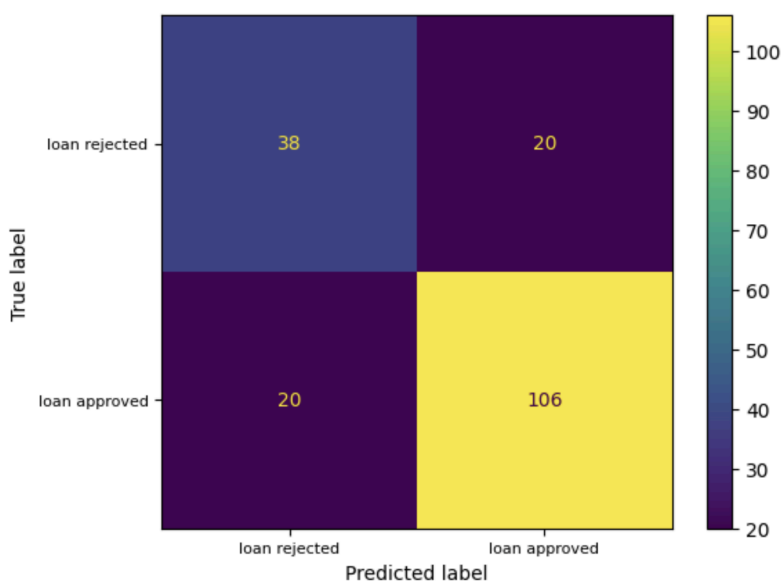
Using these parameters, the model achieved the following result on the test dataset:
Accuracy: 0.78

```
Logistic Regression: Classification Report for Test Data
              precision    recall  f1-score   support

           0       0.66      0.66      0.66        58
           1       0.84      0.84      0.84       126

    accuracy                           0.78       184
   macro avg       0.75      0.75      0.75       184
weighted avg       0.78      0.78      0.78       184
```

A confusion matrix was generated to evaluate how well the model distinguished between approved and rejected loan applications.



II. Random Forest

A Random Forest classifier was then trained to capture more complex, non-linear relationships within the dataset.

Random Forest is an ensemble learning method that combines multiple decision trees to

improve predictive performance and reduce overfitting.

The Random Forest model achieved the following performance metrics:

- Accuracy=0.826
- f1-score=0.817

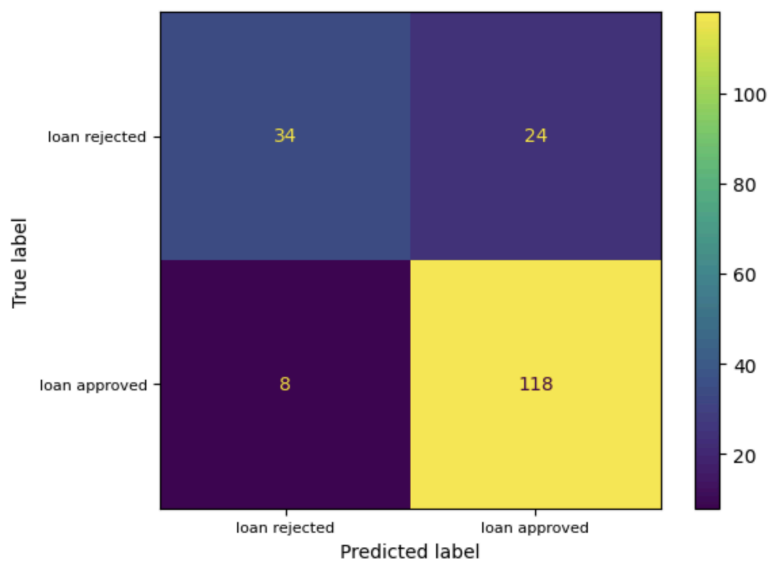
These results indicate an improvement over the Logistic Regression model, suggesting that the ensemble approach is better suited for capturing patterns within the loan eligibility data.

```
Random Forest: Classification Report for Test Data
              precision    recall  f1-score   support

           0       0.81      0.59      0.68        58
           1       0.83      0.94      0.88       126

    accuracy                           0.83       184
   macro avg       0.82      0.76      0.78       184
weighted avg       0.82      0.83      0.82       184
```

A confusion matrix was also generated to visualize classification performance across approval and rejection categories.



III. Gradient Boosting

A Gradient Boosting classifier was also evaluated as another ensemble-based approach.

Gradient Boosting builds models sequentially, where each new model attempts to correct the errors made by previous models. This approach can often achieve strong predictive performance for structured tabular datasets.

The Gradient Boosting model produced the following result:

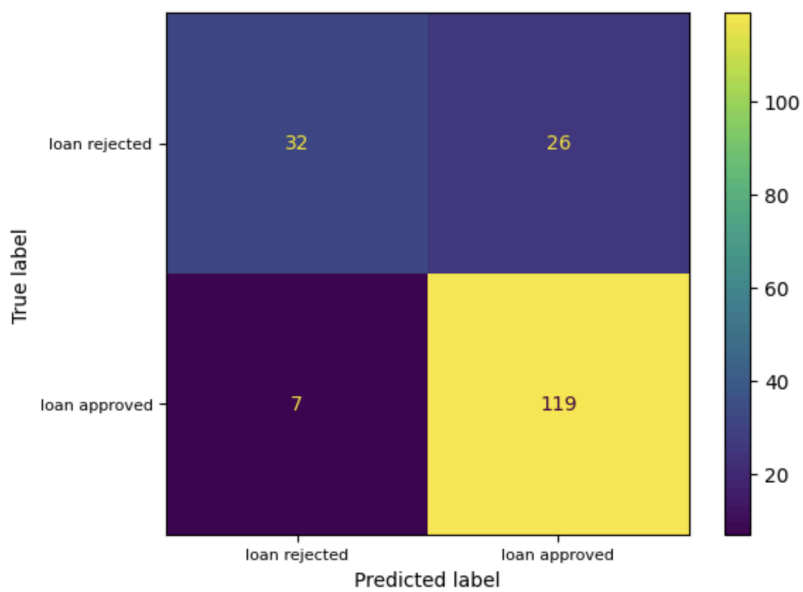
- Accuracy: 0.82

```
Gradient Boosting: Classification Report for Test Data
              precision    recall  f1-score   support

           0       0.82      0.55      0.66        58
           1       0.82      0.94      0.88       126

    accuracy                           0.82       184
   macro avg       0.82      0.75      0.77       184
weighted avg       0.82      0.82      0.81       184
```

Similar to the other models, a confusion matrix was generated to further analyze prediction accuracy across classes.



Results

Model	Accuracy
Logistic Regression	0.78
Random Forest	0.826
Gradient Boosting	0.82

The Random Forest model produced the best performance and demonstrated the strongest ability to classify loan approval outcomes.

Future Improvements

- Use larger and more diverse datasets
- Apply SMOTE or other techniques to address class imbalance
- Perform feature importance analysis
- Deploy the model as a web application for real-time predictions
- Incorporate additional financial indicators such as credit score or employment stability